



Text Classification and Naïve Bayes

The Task of Text Classification

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Is this spam?

Subject: Important notice!

From: Stanford University <newsforum@stanford.edu>

Date: October 28, 2011 12:34:16 PM PDT

To: undisclosed-recipients;;

Greats News!

You can now access the latest news by using the link below to login to Stanford University News Forum.

<http://www.123contactform.com/contact-form-StanfordNew1-236335.html>

Click on the above link to login for more information about this new exciting forum. You can also copy the above link to your browser bar and login for more information about the new services.

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Male or female author?

1. By 1925 present-day Vietnam was divided into three parts under French colonial rule. The southern region embracing Saigon and the Mekong delta was the colony of Cochinchina; the central area with its imperial capital at Hue was the protectorate of Annam...
2. Clara never failed to be astonished by the extraordinary felicity of her own name. She found it hard to trust herself to the mercy of fate, which had managed over the years to convert her greatest shame into one of her greatest assets...



Positive or negative movie review?



- unbelievably disappointing



- Full of zany characters and richly applied satire, and some great plot twists

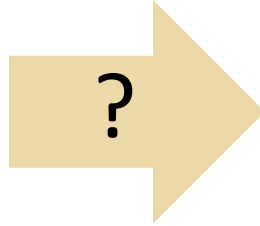


- this is the greatest screwball comedy ever filmed



- It was pathetic. The worst part about it was the boxing scenes.

MEDLINE Article



- Antagonists and Inhibitors
- Blood Supply
- Chemistry
- Drug Therapy
- Embryology
- Epidemiology
- ...



Text Classification

- Assigning subject categories, topics, or genres
- Spam detection
- Authorship identification
- Age/gender identification
- Language Identification
- Sentiment analysis
- ...



Text Classification: definition

- *Input*:
 - a document d
 - a fixed set of classes $C = \{c_1, c_2, \dots, c_J\}$
- *Output*: a predicted class $c \in C$

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Classification Methods: Hand-coded rules

- Rules based on combinations of words or other features
 - spam: black-list-address OR (“dollars” AND “have been selected”)
- Accuracy can be high
 - If rules carefully refined by expert
- But building and maintaining these rules is expensive



Classification Methods: Supervised Machine Learning

- *Input:*
 - a document d
 - a fixed set of classes $C = \{c_1, c_2, \dots, c_J\}$
 - A training set of m hand-labeled documents $(d_1, c_1), \dots, (d_m, c_m)$
- *Output:*
 - a learned classifier $\gamma: d \rightarrow c$

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Classification Methods: Supervised Machine Learning

- Any kind of classifier
 - Naïve Bayes
 - Logistic regression
 - Support-vector machines
 - k-Nearest Neighbors
- ...



The Task of Text Classification



Text Classification and Naïve Bayes

Naïve Bayes (I)



Naïve Bayes Intuition

- Simple (“naïve”) classification method based on Bayes rule
- Relies on very simple representation of document
 - Bag of words



The bag of words representation

Y (

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet.

) = C





The bag of words representation

Y (

I **love** this movie! It's **sweet**, but with **satirical** humor. The dialogue is **great** and the adventure scenes are **fun**... It manages to be **whimsical** and **romantic** while **laughing** at the conventions of the fairy tale genre. I would **recommend** it to just about anyone. I've seen it **several** times, and I'm always **happy** to see it **again** whenever I have a friend who hasn't seen it yet.

) = C





The bag of words representation: using a subset of words

Y (

x love xxxxxxxxxxxxxxxxxxxx sweet
 xxxxxxxx satirical xxxxxxxxxxxx
 xxxxxxxxxxxx great xxxxxxxx
 xxxxxxxxxxxxxxxxxxxxxxxxxxxx fun xxxx
 xxxxxxxxxxxxxxxxxxxx whimsical xxxx
 romantic xxxx laughing
 xx
 xxxxxxxxxxxxxxxxxxxx recommend xxxxxx
 xx
 xx several xxxxxxxxxxxxxxxxxxxxxxxxxxxx
 xxxxxx happy xxxxxxxxxxxx again
 xx
 xx

) = C





The bag of words representation

Y (

great	2
love	2
recommend	1
laugh	1
happy	1
...	...

) = C

thumbs up
thumbs down



Bag of words for document classification

Test
document

parser
language
label
translation
...

?

Machine
Learning

learning
training
algorithm
shrinkage
network...

NLP

parser
tag
training
translation
language...

Garbage
Collection

garbage
collection
memory
optimization
region...

Planning

planning
temporal
reasoning
plan
language...

GUI

...



Text Classification and Naïve Bayes

Formalizing the Naïve Bayes Classifier



Bayes' Rule Applied to Documents and Classes

- For a document *d* and a class *c*

$$P(c \mid d) = \frac{P(d \mid c)P(c)}{P(d)}$$



Naïve Bayes Classifier (I)

$$c_{MAP} = \operatorname{argmax}_{c \in C} P(c \mid d)$$

MAP is “maximum a posteriori” = most likely class

$$= \operatorname{argmax}_{c \in C} \frac{P(d \mid c)P(c)}{P(d)}$$

Bayes Rule

$$= \operatorname{argmax}_{c \in C} P(d \mid c)P(c)$$

Dropping the denominator



Naïve Bayes Classifier (II)

$$c_{MAP} = \operatorname{argmax}_{c \in C} P(d \mid c)P(c)$$

$$= \operatorname{argmax}_{c \in C} P(x_1, x_2, \dots, x_n \mid c)P(c)$$

Document d
represented as
features
 $x_1 \dots x_n$



Multinomial Naïve Bayes Independence Assumptions

$$P(x_1, x_2, \dots, x_n \mid c)$$

- **Bag of Words assumption:** Assume position doesn't matter
- **Conditional Independence:** Assume the feature probabilities $P(x_i \mid c_j)$ are independent given the class c .

$$P(x_1, \dots, x_n \mid c) = P(x_1 \mid c) \cdot P(x_2 \mid c) \cdot P(x_3 \mid c) \cdot \dots \cdot P(x_n \mid c)$$



Multinomial Naïve Bayes Classifier

$$c_{MAP} = \operatorname{argmax}_{c \in C} P(x_1, x_2, \dots, x_n | c) P(c)$$

$$c_{NB} = \operatorname{argmax}_{c \in C} P(c_j) \prod_{x \in X} P(x | c)$$



Learning the Multinomial Naïve Bayes Model

- First attempt: maximum likelihood estimates
 - simply use the frequencies in the data

$$\hat{P}(c_j) = \frac{\text{doccount}(C = c_j)}{N_{doc}}$$

$$\hat{P}(w_i | c_j) = \frac{\text{count}(w_i, c_j)}{\sum_{w \in V} \text{count}(w, c_j)}$$



Parameter estimation

$$\hat{P}(w_i | c_j) = \frac{\text{count}(w_i, c_j)}{\sum_{w \in V} \text{count}(w, c_j)}$$

fraction of times word w_i appears
among all words in documents of topic c_j

- Create mega-document for topic j by concatenating all docs in this topic
 - Use frequency of w in mega-document



Problem with Maximum Likelihood

- What if we have seen no training documents with the word ***fantastic*** and classified in the topic **positive** (***thumbs-up***)?

$$\hat{P}(\text{"fantastic"} \mid \text{positive}) = \frac{\text{count}(\text{"fantastic"}, \text{positive})}{\sum_{w \in V} \text{count}(w, \text{positive})} = 0$$

- Zero probabilities cannot be conditioned away, no matter the other evidence!

$$c_{MAP} = \operatorname{argmax}_c \hat{P}(c) \prod_i \hat{P}(x_i \mid c)$$



Laplace (add-1) smoothing for Naïve Bayes

$$\begin{aligned}\hat{P}(w_i | c) &= \frac{\text{count}(w_i, c) + 1}{\sum_{w \in V} (\text{count}(w, c) + 1)} \\ &= \frac{\text{count}(w_i, c) + 1}{\left(\sum_{w \in V} \text{count}(w, c) \right) + |V|}\end{aligned}$$



Multinomial Naïve Bayes: Learning

- From training corpus, extract *Vocabulary*
- Calculate $P(c_j)$ terms
 - For each c_j in C do
 - $docs_j \leftarrow$ all docs with class $= c_j$
$$P(c_j) \leftarrow \frac{|docs_j|}{|\text{total \# documents}|}$$
- Calculate $P(w_k | c_j)$ terms
 - $Text_j \leftarrow$ single doc containing all $docs_j$
 - For each word w_k in *Vocabulary*
 - $n_k \leftarrow$ # of occurrences of w_k in $Text_j$
$$P(w_k | c_j) \leftarrow \frac{n_k + \alpha}{n + \alpha |Vocabulary|}$$



$$\hat{P}(c) = \frac{N_c}{N}$$

$$\hat{P}(w|c) = \frac{\text{count}(w,c)+1}{\text{count}(c)+|V|}$$

	Doc	Words	Class
Training	1	Chinese Beijing Chinese	c
	2	Chinese Chinese Shanghai	c
	3	Chinese Macao	c
	4	Tokyo Japan Chinese	j
Test	5	Chinese Chinese Chinese Tokyo Japan	?

Priors:

$$P(c) = \frac{3}{4}$$

$$P(j) = \frac{1}{4}$$

Choosing a class:

$$P(c|d5) \propto \frac{3}{4} * \left(\frac{3}{7}\right)^3 * \frac{1}{14} * \frac{1}{14} \approx 0.0003$$

Conditional Probabilities:

$$P(\text{Chinese}|c) = (5+1) / (8+6) = 6/14 = 3/7$$

$$P(\text{Tokyo}|c) = (0+1) / (8+6) = 1/14$$

$$P(\text{Japan}|c) = (0+1) / (8+6) = 1/14$$

$$P(\text{Chinese}|j) = (1+1) / (3+6) = 2/9$$

$$P(\text{Tokyo}|j) = (1+1) / (3+6) = 2/9$$

$$P(\text{Japan}|j) = (1+1) / (3+6) = 2/9$$

$$P(j|d5) \propto \frac{1}{4} * \left(\frac{2}{9}\right)^3 * \frac{2}{9} * \frac{2}{9} \approx 0.0001$$

Example of Naïve Bayes Classifier

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

A: attributes

M: mammals

N: non-mammals

$$P(A | M) = \frac{6}{7} \times \frac{6}{7} \times \frac{2}{7} \times \frac{2}{7} = 0.06$$

$$P(A | N) = \frac{1}{13} \times \frac{10}{13} \times \frac{3}{13} \times \frac{4}{13} = 0.0042$$

$$P(A | M)P(M) = 0.06 \times \frac{7}{20} = 0.021$$

$$P(A | N)P(N) = 0.004 \times \frac{13}{20} = 0.0027$$

Give Birth	Can Fly	Live in Water	Have Legs	Class
yes	no	yes	no	?

$$P(A|M)P(M) >$$

$$P(A|N)P(N)$$

=> Mammals